

29. 290A Exam, Fall 2017

Three-hour exam. Do as many questions as you can. Each is worth 4 marks. Please write clear maths and clear English which could be understood by one of your fellow students! Include as much detail as is appropriate; you can use standard results and theorems in your answers provided you refer to them clearly.

1. Let X be the subspace of the unit cube I^3 consisting of the union of the six boundary faces and the four diagonal arc segments of length $\sqrt{3}$. Compute $\pi_1(X)$.
2. Let X be a topological space. Its *suspension* ΣX is the quotient of $X \times I$ obtained by collapsing $X \times \{0\}$ to a point and $X \times \{1\}$ to a (different) point. Compute $\pi_1(\Sigma X)$.
3. Let $X = S^1 \times S^1$, and let $A \subseteq X$ be a subspace which is homeomorphic to $S^1 \vee S^1$. Show that there is no retraction from X to A .
4. Show that if $f : S^2 \rightarrow S^1$ then there exists a point $x \in S^2$ such that $f(-x) = f(x)$.
5. Up to equivalence, how many path-connected 2-fold covering spaces does $S^1 \vee \mathbb{R}P^2$ have? Describe the total spaces of these coverings.
6. Let $G = \langle a, b \rangle$ be the free group on two symbols, and let $H \leq G$ be the subgroup generated by the elements $aba^{-1}b^{-1}, a^2ba^{-2}b^{-1}, a^{-1}bab^{-1}$ and $aba^{-2}b^{-1}a$. To what well-known group is it isomorphic?